

PROJECT DOCUMENT

FITBUDDY

AI FITNESS PLAN GENERATOR USING GEMINI MODELS

TEAM PROJECT

PROJECT PURPOSE

FitBuddy is designed to demonstrate how generative AI can be used to create a structured, personalized fitness-oriented response from general user goals and preferences.

Academic Project

Bachelor of Computer Applications (BCA)

Academic Year: 2026–2027

CERTIFICATE

This is to certify that the project entitled “**FitBuddy – AI Fitness Plan Generator using Gemini Models**” is a team project carried out as part of the academic project requirements.

The project follows the prescribed phase-wise project development process and documents the idea, requirements, design, planning, development, testing, documentation and demonstration activities.

Project Guide	_____
Head of Department	_____
Project Coordinator	_____

ACKNOWLEDGEMENT

We express our sincere gratitude to our project guide, faculty members and trainers for their guidance and support throughout the development of this team project. Their suggestions helped us understand the importance of following a clear, phase-wise software project process.

We also thank our college for providing the academic environment and resources required to complete FitBuddy and for supporting our brainstorming, preparation, testing and final demonstration.

PROJECT OVERVIEW

FitBuddy is an AI-powered fitness plan generator concept that uses Gemini models to transform general user preferences and goals into a structured response. The project focuses on demonstrating the practical use of generative AI in a simple and understandable student team project.

The main problem addressed by FitBuddy is the difficulty of turning a person's general fitness goal, preferred activities and available routine into a clearly organized starting plan. Instead of manually preparing every possible response, the system uses a Gemini model to generate a response from structured user information.

The project is not intended to diagnose medical conditions or replace professional advice. It is an academic demonstration of how an AI service can be integrated into an application to produce useful, organized information.

PROJECT OBJECTIVES

- Create a simple and user-friendly fitness planning experience.
- Use Gemini models to generate structured responses from user inputs.
- Demonstrate the complete project development process.
- Maintain the project phase-wise as required for submission.
- Prepare clear documentation and a complete demonstration of the final project.

PROJECT FLOW

User Goal & Preferences → FitBuddy → Gemini Model → Structured AI Response

The following eight highlighted phases form the main development base of this team project.

1. BRAINSTORMING & IDEATION PHASE

WHAT IS THIS PHASE?

This is the stage where the project idea is discussed, different possible solutions are considered and the final project concept is selected.

WHY IS IT IMPORTANT?

A clear idea is necessary before starting the project. This phase helps define the actual problem, the target use case and what the project should finally achieve.

WHAT PROBLEM DOES FITBUDDY ADDRESS?

Many people know that they want to improve their fitness but may not know how to organize their general goal, preferred activities and available routine into a clear starting structure. FitBuddy addresses this information-organization problem by using generative AI to produce an organized response.

WHAT WE DECIDED TO DO

- Create an AI-based fitness plan generator named FitBuddy.
- Use Gemini models as the generative AI component.
- Collect general user preferences and goals.
- Convert the information into a structured request.
- Display the generated result in a clear format.

FINAL IDEA

FitBuddy was selected as a practical team project because it combines a real-world use case with current AI technology while keeping the project scope understandable and demonstrable.

2. REQUIREMENT ANALYSIS PHASE

WHAT IS THIS PHASE?

This phase identifies exactly what FitBuddy should do, what information it needs and what the user should see at each stage.

MAIN REQUIREMENTS

- **User Input:** Collect the project-defined general fitness goals and preferences.
- **Validation:** Check required information before generating a response.
- **AI Generation:** Convert collected information into a suitable prompt for the Gemini model.
- **Result Display:** Present the generated response in a readable and organized manner.
- **Error Handling:** Show a clear message if information is missing or the AI service cannot respond.
- **Reset:** Allow the user to start a new request.

WHAT PROBLEM DOES THIS PHASE SOLVE?

Without requirement analysis, a project can become unclear and difficult to manage. This phase prevents unnecessary features from being added and gives the team a clear understanding of what must be completed.

EXPECTED RESULT

At the end of this phase, FitBuddy has a clear list of required functions, expected behaviour and project boundaries.

3. PROJECT DESIGN PHASE

WHAT IS THIS PHASE?

The design phase decides how FitBuddy will work from the user's first interaction to the final AI-generated response.

BASIC SYSTEM DESIGN

PART	WHAT IT DOES
User Interface	Collects the required information and displays the result.
Application Process	Checks information, prepares the AI request and handles the response.
Gemini Model	Generates the response based on the structured request.

USER FLOW

Enter Goal & Preferences → Submit → Information Check → Gemini Processing → Generated Fitness-Oriented Response

DESIGN GOAL

The interface should be simple, clear and easy to understand. The user should know what information is required, when the request is being processed and where the final response appears.

SAFETY CONSIDERATION

The project should clearly present the output as general informational guidance and should not claim to provide medical diagnosis or treatment.

4. PROJECT PLANNING PHASE

WHAT IS THIS PHASE?

Planning divides the project into manageable activities and connects each activity with the evidence required for submission.

PHASE	WHAT WE DO	OUTPUT
Brainstorming & Ideation	Finalize the project idea	Problem + solution
Requirement Analysis	Define what the system must do	Requirement list
Project Design	Decide the structure and user flow	Design description
Project Planning	Set activities and milestones	Project schedule
Project Development	Build the working project	Working prototype
Project Testing	Check expected and error cases	Testing evidence
Project Documentation	Prepare report and repository	Final documentation
Project Demonstration	Show the completed project	Demo video

WHAT PROBLEM DOES PLANNING SOLVE?

Planning prevents last-minute confusion. It makes the work easier to track and ensures that all eight required phases are covered before submission.

WHAT WE WILL MAINTAIN

- Clear project milestones.
- Phase-wise GitHub updates.
- Testing evidence and screenshots.
- Final documentation and demonstration video.

5. PROJECT DEVELOPMENT PHASE

WHAT IS THIS PHASE?

This is the stage where the planned FitBuddy concept is turned into a working application.

WHAT HAPPENS HERE?

- The user interface is prepared according to the approved design.
- The required input information is collected.
- The input is checked before processing.
- A structured request is prepared for the Gemini model.
- The AI-generated response is received and displayed.
- The application provides understandable messages when something goes wrong.

WHAT PROBLEM DOES DEVELOPMENT SOLVE?

The earlier phases describe what FitBuddy should be. Development turns those ideas and designs into an actual working project that can be tested and demonstrated.

EXPECTED FINAL OUTPUT

A user should be able to open FitBuddy, enter the required information, request an AI-generated response and view the result in an organized format.

PROJECT REPOSITORY

The project files and documentation should be maintained in the required public GitHub repository. Private API credentials must never be exposed in a public repository.

6. PROJECT TESTING PHASE

WHAT IS THIS PHASE?

Testing checks whether FitBuddy works correctly under normal conditions and also handles common problems properly.

WHAT WE CHECK

- Valid information produces a generated response.
- Missing required information produces a clear validation message.
- A new request can be made after a previous request.
- Temporary service or network problems are handled clearly.
- The reset action clears the previous information.
- The interface remains understandable on a phone-sized screen.

WHY IS TESTING IMPORTANT?

Testing helps identify mistakes before the final demonstration. It also provides evidence that the project was actually checked rather than only prepared as a concept.

7. PROJECT DOCUMENTATION PHASE

Documentation records the complete story of the project: why it was created, what problem it addresses, how it was designed, what was developed, how it was tested and how it will be demonstrated.

DOCUMENTATION SHOULD INCLUDE

- Project overview and objective.
- Details of all eight development phases.
- Screenshots and relevant project evidence.
- GitHub repository information.
- Testing results and known limitations.
- Final demonstration details.

8. PROJECT DEMONSTRATION PHASE

WHAT IS THIS PHASE?

The demonstration phase shows the completed FitBuddy project from beginning to end so that the reviewer can understand what was created and how it works.

DEMONSTRATION FLOW

- Introduce the project title.
- Explain the problem and the reason for choosing the project.
- Show the main user interface.
- Enter sample project-defined information.
- Generate the AI response and show the result.
- Demonstrate at least one validation or error-handling case.
- Show the phase-wise GitHub repository.
- Conclude with the project outcome and future scope.

FUTURE ENHANCEMENTS

- Multilingual user interface.
- Improved response formatting and accessibility.
- Progress tracking using appropriate non-sensitive information.
- More configurable AI response templates.
- Additional validation and safety-focused response controls.

CONCLUSION

FitBuddy demonstrates how generative AI can be applied to a practical student team project. The eight required phases provide a clear development path from brainstorming to final demonstration. The project focuses on solving the problem of organizing general fitness goals and preferences into a structured AI-generated response while keeping the application understandable, documented and demonstrable.

REFERENCES

- Google Gemini and official Gemini API developer resources.
- GitHub documentation for repositories and version control.
- Project Development Session guidelines supplied by the training team.
- General software engineering principles for project requirements, design, development and testing.